LAB WRITEUP 05 - DNS IN DETAIL (TRYHACKME)

Room	DNS in Detail
Platform	TryHackMe - `https://tryhackme.com/room/dnsindetail`
Path	Pre Security / Cyber Security 101 - How the Web Works
Difficulty	Easy
Time to complete	About 45 minutes
Tasks	5

Objective

Understand what DNS is, how the domain hierarchy is built, which record types exist, the exact journey of a DNS request from a browser to an authoritative nameserver, and how to run real DNS queries.

Concepts covered

- Purpose of DNS and why IP addresses alone are unusable for humans
- Domain hierarchy: TLD, second-level domain, subdomain, and their length limits
- Record types: A, AAAA, CNAME, MX, TXT
- Recursive vs authoritative nameservers, caching and TTL
- Practical queries with the room's DNS simulation

Task 1 - What is DNS?

DNS (Domain Name System) translates human-friendly domain names into the IP addresses that machines actually route to. It exists because remembering `104.26.10.229` is impractical, while remembering `example.com` is not.

Question	Answer
What does DNS stand for?	Domain Name System

Task 2 - Domain Hierarchy

A domain name is read from right to left:

- TLD (Top Level Domain) - the right-most label, e.g. `.com`. Two kinds exist: gTLD (generic, such as `.com`, `.org`, `.edu`, `.gov`) and ccTLD (country code, such as `.uk`, `.br`, `.in`).
- Second-level domain - the label the organisation registers, e.g. `example` in `example.com`. Maximum 63 characters; letters `a-z`, digits `0-9` and hyphens only.
- Subdomain - anything to the left of the second-level domain, e.g. `shop` in `shop.example.com`. Subdomains can be stacked (`a.b.example.com`); each label has the same 63 character limit, and the whole name may not exceed 253 characters.

Question	Answer	Reasoning
Maximum length of a subdomain	63	One label of a DNS name is limited to 63 characters.

Question	Answer	Reasoning
Which character cannot be used in a subdomain: `3`, `b`, `_`, `-`?	_	Only letters, digits and hyphens are permitted; the underscore is not.
Maximum length of a domain name	253	The total dotted name, including all dots and the TLD, may not exceed 253 characters.
What type of TLD is `.co.uk`?	ccTLD	`.uk` is a country-code TLD, and `.co.uk` is its commercial second-level registration space.

Task 3 - DNS Record Types

Record	Points to	Example use
A	An IPv4 address	`www.example.com -> 10.10.10.10`
AAAA	An IPv6 address	`example.com -> 2606:4700:20::681a:be`
CNAME	Another domain name (an alias)	`shop.example.com -> shops.myshopify.com`; the resolver then resolves the target
MX	The mail servers for a domain, each with a priority number	Lower priority = tried first, so a backup mail server can be configured
TXT	Free text	SPF/DKIM/DMARC records, domain-ownership verification - and a favourite channel for DNS tunnelling

Question	Answer
Which record type advises where to send email?	MX
Which record type handles IPv6 addresses?	AAAA

Task 4 - Making a Request

The full path of a DNS lookup:

1. Local cache first. The client checks its own cache; if the name was resolved recently and the TTL has not expired, the answer is used immediately and nothing leaves the machine.
2. Recursive DNS server. Otherwise the stub resolver sends a recursive query to the recursive resolver, usually supplied by the ISP (or a public resolver such as `1.1.1.1`). This server keeps its own cache, which is why popular sites are almost always answered from cache.
3. Root servers. On a cache miss the resolver asks a root server, which returns a referral to the correct TLD server (for `www.google.com`, the `.com` TLD).
4. TLD server. The TLD server returns a referral to the domain's authoritative nameservers (for example `kip.ns.cloudflare.com`).
5. Authoritative nameserver. This server holds the zone records and returns the real answer, which the recursive resolver caches - for the number of seconds given by the TTL - and forwards to the client. The browser then connects to that IP address.

Question	Answer	Reasoning
Which field specifies how long a DNS record should be cached?	TTL	Time To Live, in seconds: how long any cache may reuse the record before asking again.
Which type of DNS server is usually provided by your ISP?	Recursive	It performs the lookups on the client's behalf and caches the results.
Which type of server holds all the records for a domain?	Authoritative	It owns the zone, so it is the only source of the definitive answer.

Task 5 - Practical

The room provides a small web application that simulates a DNS server, and the records must be read from the query results.

Question	Answer
CNAME of `shop.website.thm`	shops.myshopify.com
Value of the TXT record of `website.thm`	THM{7012BBA60997F35A9516C2E16D2944FF}
Numerical priority value for the MX record	30
IP address for the A record of `www.website.thm`	10.10.10.10

Each answer comes from selecting the record type in the simulation (or querying it with `nslookup -type=<record> <name>`) and reading the returned value. In the real world the equivalent is:

```
nslookup -type=CNAME shop.website.thm      -> alias target
nslookup -type=TXT   website.thm           -> free-text record
nslookup -type=MX    website.thm           -> "MX preference = 30, mail exchanger = ..."
nslookup -type=A     www.website.thm       -> "Address: 10.10.10.10"
```

Key takeaways

1. DNS is a distributed, hierarchical, cached database - root, TLD, authoritative - not one server.
2. Caching at every layer plus TTL is what makes the internet's name system scalable; it also explains why an old malicious answer can still be in a cache.
3. Know the record types: A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail, with priority), TXT (text - and a common exfiltration channel).
4. Length and character limits matter: 63 characters per label, 253 for the whole name, and no underscores.
5. For a SOC analyst, DNS logs are one of the richest evidence sources because DNS is rarely encrypted and almost all connections start with a lookup.